

Ultimate Nutrition

Beta K

A recent study illustrated that supplementation of Beta-Alanine & Creatine increased maximal power output. It was concluded that it might be as a result of improved hydrogen ion buffering by elevated Carnosine concentrations. The use of Beta-Alanine to increase muscle Carnosine offers a way to increase anaerobic exercise potential and, is likely to add to the effects of Creatine in some exercise settings.

First discovered in the early 1900's, Beta-Alanine and Histidine are the two components of Carnosine. Histidine is already present in large quantity within skeletal muscles, so it is Beta-Alanine that acts as the rate-limiting factor in Carnosine conversion.

Carnosine is very effective at buffering the Hydrogen ions responsible for producing the lactic acid burn. Studies have shown that increasing muscle Carnosine by supplementing Beta-Alanine may delay fatigue and improve the muscular aspects of athletic performance.

Research also indicates that Beta-Alanine increases lactate threshold, improves the ability to maintain maximal power output during high-intensity exercise and decreases neuromuscular fatigue. By elevating Carnosine level, it may protect against damage to nerves, letting them fire at a faster rate than if they are injured. For example, instead of ones body operating at 80% the day after exercise, Carnosine may help it perform closer to the best possible levels. This is mainly useful for athletes who are constantly using the same muscles, without the ability to take a break and recover for a few days.

The effects of creatine plus beta-alanine on strength, power, body composition, and endocrine changes were examined during a 10-wk resistance-training program in collegiate football players. Results of this study demonstrate the efficacy of creatine plus beta-alanine on strength performance. Creatine plus beta-alanine supplementation appeared to have the greatest effect on lean tissue accretion and body fat composition.

Creatine Monohydrate has been proven through decades of research to enhance strength, endurance, and athletic performance. However some athletes have noted side effects including nausea, stomach cramps, bloating, and a host of other unwanted reactions.

SELECTED REFERENCES:

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FREQUENTLY ASKED QUESTIONS:

1. What is the tingling feeling I am experiencing?

Beta-Alanine is known to cause a niacin-like skin tingling reaction within minutes after it is consumed. This is perfectly normal and lessens with continued Beta-Alanine supplementation. It has best been described as the skin having begun to tingle or itch in certain spots. This is a normal body reaction to Beta-Alanine consumption.

2. Shouldn't we take extra histidine along with Beta-Alanine since histidine is a component of carnosine?

No, because histidine is already present in high concentrations in muscle, while Beta-Alanine is only present only in small amounts. Researchers have determined that it is beta-alanine that drives carnosine synthesis, not histidine. For the majority of healthy people, eating a well balance diet, histidine deficiency is very uncommon and no extra is needed to increase carnosine, just Beta-Alanine.